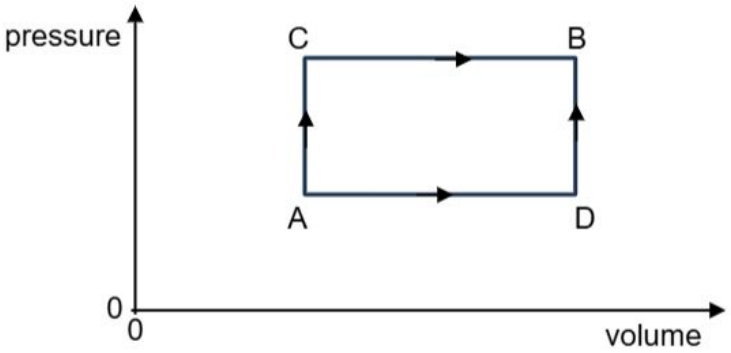
18	A fixed mass of gas undergoes changes of pressure and volume as shown.

[Turn over



When the gas is taken from state A to state B by the stages AC and CB, 6 J of work is done by it and there is a thermal transfer of 16 J of energy into it.

When the same resultant change is achieved by stages AD and DB, 2 J of work is done by the gas.

In this case, what is the thermal transfer of energy?

	A	8 J out of the gas		
	B	12 J into the gas		
	C	16 J into the gas		
	D	20 J into the gas		

L2 **Answer: B**

Same initial & final states in both pathways, hence internal energy gain is the same for both pathways, since internal energy is a state properly (path-independent).

Stage AC and CB:
Internal energy gained = Q + ΔW = 16 + (– 6) = 10 J

Stage AD and DB:
Internal energy gained = Q + ΔW
10 = Q + (– 2)
Q = +12 J
12 J of heat supplied